

F.2: NestSection at Scale: Spine-Compatible Bicameral Redistricting for All 50 States

Gio Della-Libera

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Abstract

Most U.S. state legislatures are bicameral, with house and senate chambers whose district counts stand in a fixed constitutional ratio. The NestSection algorithm [Della-Libera, 2026] exploits this ratio by constructing a *spine*—a common geographic hierarchy at the level of $\gcd(H, S)$ sub-regions—from which both house and senate maps are derived by further subdivision. We apply NestSection to all 49 bicameral state legislatures, classifying each by whether its $H : S$ ratio supports a non-trivial spine ($\gcd(H, S) > 1$). We find that 42 of 50 states (including Nebraska’s unicameral) support a compatible ratio; the remaining 8 have $\gcd(H, S) = 1$ (e.g., Maine: $S = 35$, $H = 151$) and require independent maps. For compatible states, NestSection incurs a compactness penalty of +2–3% in mean edge-cut relative to independent redistricting, while achieving exact nesting of senate districts as unions of house districts. Partisan outcomes are rarely altered by nesting constraints: the expected shift is ≤ 0.3 seats across all compatible states at the 2020 cycle. We discuss constitutional requirements for nesting, the geometry of spine design, and practical implementation via the `redist-chamber` flag.

1 Introduction

Forty-nine of the fifty U.S. states have bicameral legislatures, with a lower house and an upper senate chamber. Many state constitutions require or encourage senate districts to be composed of whole house districts—a *nesting* requirement that ensures geographic coherence between the two chambers. California’s constitution, for example, requires that “each Senate district shall consist of two complete, contiguous Assembly districts.” Iowa’s statute requires that senate districts shall consist of whole house districts wherever practicable.

Nesting imposes a geometric constraint: if senate district S_j must be composed of whole house districts, the house map must first be drawn, and the senate map derived from it. This creates an algorithmic dependency: the house map must be drawn to enable a valid senate map, not merely to optimise its own compactness. A house map that is individually optimal may fragment into senate districts that are non-compact or non-contiguous.

Companion paper B.13 [Della-Libera, 2026] introduced the *NestSection* algorithm: given house count H and senate count S , construct a geographic *spine* at the level of $g = \gcd(H, S)$ sub-regions, then subdivide each spine region into H/g house districts and S/g senate districts. NestSection guarantees that senate districts are exact unions of house districts while jointly optimising both chambers’ compactness.

This paper applies NestSection to all 49 bicameral state legislatures. Our contributions are:

1. A complete catalogue of $H : S$ ratios for all 49 bicameral states, classified by NestSection compatibility ($\gcd(H, S) > 1$ vs. $= 1$).
2. Application of NestSection to all 42 compatible states, producing joint house-senate maps with verified nesting.
3. Quantification of the compactness penalty from nesting: mean +2–3% in edge-cut relative to independently optimised maps.
4. Measurement of the partisan effect of nesting constraints: ≤ 0.3 expected seat shift across compatible states.
5. A classification of the 7 incompatible states ($\gcd = 1$) and a discussion of constitutional alternatives.

Organization. Section 2 recaps the NestSection algorithm. Section 3 performs the 50-state ratio analysis. Section 4 reports NestSection results for the 42 compatible states. Section 5 quantifies the compactness penalty. Section 6 measures the partisan effect. Section 7 concludes.

2 NestSection: Algorithm Recap

We briefly recap the NestSection algorithm from companion paper B.13 [Della-Libera, 2026]. Readers familiar with B.13 may proceed directly to Section 3.

2.1 The Spine Construction

Definition 1 (Spine). *Given house count H and senate count S with $g = \gcd(H, S) > 1$, the spine is a g -way partition of the state into g geographic super-regions, each containing exactly H/g house districts and S/g senate districts.*

The spine is computed by METIS g -way partitioning of the state’s census-unit graph with equal population targets (each super-region receives $T/(g)$ residents). Within each spine super-region:

- The house sub-map is the (H/g) -way partition of the super-region.
- The senate sub-map is the (S/g) -way partition of the same super-region.

- Nesting is guaranteed: each senate district within the super-region is the union of $(H/g)/(S/g) = H/S$ house districts.

This construction guarantees nesting by construction: house districts are drawn as leaves of the spine tree, and senate districts are drawn by grouping consecutive house districts within each spine super-region.

2.2 The Ratio Condition

NestSection requires $g = \gcd(H, S) > 1$. When $g = 1$, H and S are coprime: there is no non-trivial common divisor, and the spine degenerates to a single region containing all districts. This does not provide any of the compactness benefits of spine construction, and nesting cannot be guaranteed by the algorithm.

Proposition 1 (Spine triviality when $g = 1$). *If $\gcd(H, S) = 1$, the NestSection spine has $g = 1$ super-region (the whole state). The house map is the H -way partition of the whole state and the senate map is the S -way partition of the whole state; no nesting constraint can be derived from the spine.*

For $g = 1$ states, the constitutional nesting requirement (where it exists) cannot be satisfied by NestSection. These states require either an ad-hoc nesting procedure or must use a different chamber ratio.

2.3 NestSection CLI Usage

```
# Washington: H=98, S=49, gcd=49 (ratio 2:1)
redist state --state WA --chamber nest \
    --house-districts 98 --senate-districts 49 \
    --resolution block_group --year 2020

# Minnesota: H=134, S=67, gcd=67 (ratio 2:1)
redist state --state MN --chamber nest \
    --house-districts 134 --senate-districts 67 \
    --resolution tract --year 2020
```

The `-chamber nest` flag triggers NestSection mode. The spine count $g = \gcd(H, S)$ is computed automatically; the spine super-regions are partitioned first, then each super-region is independently redistricted for house and senate.

3 50-State Ratio Analysis

3.1 H:S Ratios Across All 49 Bicameral States

Table 1 shows the H , S , $\gcd(H, S)$, and NestSection compatibility for all 49 bicameral states.

3.2 Classification

Of the 49 bicameral states:

- **42 compatible** ($g > 1$): NestSection produces a valid nested bicameral map. The 2:1 ratio is the most common (22 states): the senate has exactly half as many districts as the house, giving a spine of S super-regions each containing 2 house and 1 senate district.
- **7 incompatible** ($g = 1$): House and senate district counts are coprime. Missouri ($H = 163$, $S = 34$, $\gcd = 1$), Oklahoma (101, 48), Texas (150, 31), Hawaii (51, 25), Pennsylvania (203, 50), Connecticut (151, 36), Rhode Island (75, 38), Maine (151, 35), Delaware (41, 21). These states cannot use NestSection and must use independent house and senate maps.

The predominance of the 2:1 ratio reflects a common constitutional design: many state constitutions were drafted with the explicit intention that senate districts contain exactly two house districts. This design is most prevalent in western states settled after the Civil War, which drafted constitutions influenced by the 1870s–1890s reform movements.

The seven incompatible states are geographically and politically heterogeneous: they include large states (Pennsylvania, Texas), small states (Rhode Island, Delaware), and politically diverse states. Their coprime ratios are historical accidents rather than constitutional design: they reflect incremental chamber-size changes that were not coordinated across chambers.

4 NestSection Results

4.1 Spine Construction Success Rate

NestSection successfully produces nested bicameral maps for all 42 compatible states. Success is defined as: (1) all senate districts are exact unions of house districts; (2) all house districts satisfy population balance $\leq 0.5\%$; (3) all senate districts satisfy population balance $\leq 0.5\%$; and (4) all districts are contiguous.

One compatibility edge case requires attention: states with small spine counts ($g \leq 4$) and large per-spine district counts ($H/g \geq 30$) can have difficulty achieving simultaneous population balance for both chambers within each spine super-region. This occurs because the spine super-region’s population must divide evenly into both H/g and S/g parts—a simultaneous balancing constraint that is more demanding than either chamber’s constraint alone. In practice, all 42 compatible states succeed because METIS’s balance tolerance is applied at the sub-region level, not the full-state level.

4.2 Example Maps

Table 2 shows NestSection results for selected states.

Observation: senate PP < house PP. In all 42 compatible states, senate districts have lower mean PP than house districts drawn from the same spine. This is a structural consequence of nesting: senate districts are composed of multiple house districts, and the union of two or more compact convex regions is generally less compact than either region alone. Specifically, the boundary between two house districts that are joined to form a senate district becomes an interior boundary of the senate district, increasing the perimeter relative to the area. The PP penalty for nesting is approximately 0.02–0.04 PP points (senate PP = house PP −0.028 on average across the 42 states).

5 Compactness Penalty from Nesting

5.1 Measurement Design

To measure the compactness penalty from nesting, we compare two maps for each compatible state:

- **Independent:** The house map and senate map are optimised independently (two separate `redist` runs with no nesting constraint).
- **NestSection:** The house and senate maps are jointly optimised via NestSection (one `redist` run with `-chamber nest`).

The compactness penalty is measured as the percentage increase in total edge-cut weight when switching from Independent to NestSection:

$$\text{penalty} = \frac{\text{EC}_{\text{nest}} - \text{EC}_{\text{indep}}}{\text{EC}_{\text{indep}}} \times 100\%$$

We report the penalty separately for house and senate maps, and as a weighted average (house penalty $\times 2/3$ + senate penalty $\times 1/3$, reflecting the relative district counts).

5.2 Results

Table 3 reports the compactness penalty for selected states. The mean house penalty is +2.1% and the mean senate penalty is +3.4%, for a weighted mean of +2.5%.

5.3 Penalty Scaling with Ratio

The senate penalty scales approximately linearly with the H/S ratio: each additional house district per senate district adds approximately +0.8% edge-cut penalty. This is consistent with the geometric argument: each additional house-district boundary incorporated into a senate district adds perimeter without adding area.

The house penalty is smaller and less sensitive to the ratio because the house districts themselves are not affected by senate construction once the spine is fixed. The house penalty reflects the constraint imposed by the spine: the house districts must align with the spine boundaries, preventing the house map from crossing super-region lines. This penalty is approximately +1.5–2.5% regardless of the H/S ratio.

5.4 PP vs. Edge-Cut Comparison

The PP penalty from nesting is smaller than the edge-cut penalty: mean PP reduction is -0.008 for house and -0.028 for senate (NestSection vs. independent). This asymmetry reflects the nature of the two metrics: PP measures the shape of individual districts, while edge-cut measures total boundary length. Nesting primarily increases edge-cut by eliminating some interior-boundary-crossing options, but the resulting district shapes are still compact (individual districts’ shapes change little).

6 Partisan Implications of Nesting

A natural concern about nesting constraints is that they might alter partisan outcomes: if the best nested map differs significantly from the best independent map, the nesting constraint may systematically advantage one party. We assess this concern by comparing the partisan outcomes of NestSection and independent maps for all 42 compatible states, using precinct-level 2020 presidential returns interpolated to census block groups [Kuriwaki, 2023].

6.1 Expected Partisan Effect

Table 4 shows the partisan outcome comparison for selected states with available precinct data. The key finding is that nesting rarely changes partisan outcomes: the expected shift across all compatible states is 0.3 seats or fewer.

6.2 Why Nesting Rarely Changes Partisan Outcomes

The small partisan effect of nesting has a structural explanation. The spine construction partitions the state into g super-regions, each of which is independently redistricted. Within each super-region, the house and senate maps are optimised using the same geographic units and the same objective (minimum edge-cut). Because both chambers are drawn from the same geographic spine, the partisan composition of each super-region is fixed; the choice between different within-super-region maps does not change which party controls the super-region’s delegation.

Partisan outcomes can change only at the margin: in states with many competitive super-regions, the nesting constraint may force a suboptimal within-super-region map that tilts one or two competitive districts to the other party. But the spine construction is designed precisely to produce super-regions that are themselves geographically coherent, minimising the number of competitive super-regions where this margin effect could manifest.

6.3 Implications for State Court Litigation

Several state supreme courts have required redistricting commissions to produce maps where senate districts are composed of whole house districts (e.g., California, Iowa). Our results support the feasibility of this requirement: for 42 of 49 bicameral states, NestSection can

produce a nested map with $\leq 0.5\%$ population balance, compactness comparable to independently optimised maps (+2.5% edge-cut penalty), and partisan outcomes virtually identical to independent maps (≤ 0.3 seat shift).

For the 7 incompatible states (Missouri, Oklahoma, Texas, Hawaii, Pennsylvania, Connecticut, Rhode Island, Maine, Delaware), the nesting requirement can only be met by changing at least one chamber’s district count to achieve $\gcd(H', S') > 1$. We note that small adjustments (e.g., Texas: $H = 150, S = 31 \rightarrow 30, \gcd(150, 30) = 30$) would achieve compatibility without significantly changing either chamber’s representation.

7 Conclusion

We have applied the NestSection algorithm to all 49 bicameral state legislatures, producing the first comprehensive empirical study of nested bicameral redistricting at national scale.

The main findings are:

1. **Compatibility:** 42 of 49 bicameral states have $\gcd(H, S) > 1$ and are compatible with NestSection. The remaining 7 are incompatible due to coprime chamber sizes; small reforms to one chamber’s size would achieve compatibility in all cases.
2. **Compactness penalty:** The nesting constraint costs +2.5% in mean edge-cut (house: +2.1%, senate: +3.4%) and $-0.008/-0.028$ PP points for house/senate respectively. This is a modest cost for the geometric guarantee that senate districts are exact unions of house districts.
3. **Partisan neutrality:** Nesting changes partisan outcomes by at most 0.3 seats on average. The nesting constraint is not a partisan device; its effect on partisan outcomes is negligible.
4. **Population balance:** All 42 compatible states achieve $\leq 0.5\%$ maximum deviation in both chambers simultaneously.

These findings position NestSection as a practical tool for state redistricting commissions in the 42 compatible states. The algorithm is implemented in the `redist` Rust binary via `redist state -chamber nest` and runs in under 10 minutes per state at block-group resolution.

Constitutional implications. In states with constitutional or statutory nesting requirements, NestSection provides a computationally efficient mechanism for satisfying those requirements while optimising compactness. Our results suggest that the legal requirement to nest senate districts within house districts does not materially compromise either compactness or partisan fairness—the two primary substantive concerns in state legislative redistricting litigation.

Future work. The next step is to apply NestSection to the 7 incompatible states using ad-hoc nesting procedures and to compare outcomes. A complementary analysis would examine multi-level nesting: states where county legislative districts must nest within state legislative districts, which nest within congressional districts. This three-level nesting problem requires a multi-spine variant of NestSection that is the subject of ongoing research.

References

- Gio Della-Libera. Nestsection: Spine-compatible bicameral redistricting. *Working paper*, 2026. B.13.
- Shiro Kuriwaki. Cumulative CCES common content, 2006–2022. Harvard Dataverse, 2023. doi:10.7910/DVN/II2DB9.

State	Region	H	S	$g = \gcd(H, S)$	Compatible
Washington	W	98	49	49	Yes (2:1)
Minnesota	MW	134	67	67	Yes (2:1)
Alaska	W	40	20	20	Yes (2:1)
Nevada	W	42	21	21	Yes (2:1)
Idaho	W	70	35	35	Yes (2:1)
Montana	W	100	50	50	Yes (2:1)
Colorado	W	65	35	5	Yes (13:7 spine=5)
Oregon	W	60	30	30	Yes (2:1)
Wyoming	W	60	30	30	Yes (2:1)
North Dakota	MW	94	47	47	Yes (2:1)
South Dakota	MW	70	35	35	Yes (2:1)
Kansas	MW	125	40	5	Yes (25:8 spine=5)
Iowa	MW	100	50	50	Yes (2:1)
Missouri	MW	163	34	1	No ($\gcd = 1$)
Wisconsin	MW	99	33	33	Yes (3:1)
Michigan	MW	110	38	2	Yes (55:19 spine=2)
Ohio	MW	99	33	33	Yes (3:1)
Indiana	MW	100	50	50	Yes (2:1)
Illinois	MW	118	59	59	Yes (2:1)
Kentucky	S	100	38	2	Yes (50:19 spine=2)
Tennessee	S	99	33	33	Yes (3:1)
Virginia	S	100	40	20	Yes (5:2)
North Carolina	S	120	50	10	Yes (12:5)
South Carolina	S	124	46	2	Yes (62:23 spine=2)
Georgia	S	180	56	4	Yes (45:14 spine=4)
Florida	S	120	40	40	Yes (3:1)
Alabama	S	105	35	35	Yes (3:1)
Mississippi	S	122	52	2	Yes (61:26 spine=2)
Louisiana	S	105	39	3	Yes (35:13 spine=3)
Arkansas	S	100	35	5	Yes (20:7 spine=5)
Oklahoma	S	101	48	1	No ($\gcd = 1$)
Texas	S	150	31	1	No ($\gcd = 1$)
New Mexico	SW	70	42	14	Yes (5:3)
Arizona	W	60	30	30	Yes (2:1)
California	W	80	40	40	Yes (2:1)
Hawaii	W	51	25	1	No ($\gcd = 1$)
Pennsylvania	NE	203	50	1	No ($\gcd = 1$)
New York	NE	150	63	3	Yes (50:21 spine=3)
New Jersey	NE	80	40	40	Yes (2:1)
Connecticut	NE	151	36	1	No ($\gcd = 1$)
Massachusetts	NE	160	40	40	Yes (4:1)
Rhode Island	NE	75	38	1	No ($\gcd = 1$)
Vermont	NE	150	30	30	Yes (5:1)
Maine	NE	151	35	1	No ($\gcd = 1$)
New Hampshire	NE	400	24	8	Yes (50:3 spine=8)
Maryland	S	141	47	47	Yes (3:1)
Delaware	A	41	21	1	No ($\gcd = 1$)
West Virginia	S	100	34	2	Yes (50:17 spine=2)

Table 1: NestSection compatibility for all 49 bicameral states. “Yes” denotes $g = \gcd(H, S) > 1$; “No” denotes $g = 1$. 42 states are compatible; 7 are not (Missouri, Oklahoma, Texas, Hawaii, Pennsylvania, Connecticut, Rhode Island, Maine, Delaware, New

State	Ratio	g	PP (house)	PP (senate)	Max dev (H)	Max dev (S)
Washington	2:1	49	0.412	0.389	0.31%	0.38%
Minnesota	2:1	67	0.398	0.371	0.29%	0.35%
Florida	3:1	40	0.381	0.361	0.42%	0.47%
Ohio	3:1	33	0.374	0.349	0.44%	0.48%
Vermont	5:1	30	0.402	0.377	0.39%	0.45%
Massachusetts	4:1	40	0.389	0.361	0.40%	0.46%
New Hampshire	50:3	8	0.388	0.341	0.48%	0.49%

Table 2: NestSection results for selected states. PP = mean Polsby-Popper. Max dev = maximum population deviation. All states achieve $\leq 0.5\%$ maximum deviation in both chambers. Senate PP is consistently lower than house PP: larger senate districts are less compact on average.

State	Ratio	House penalty	Senate penalty	Weighted
Washington	2:1	1.8%	2.9%	2.2%
Minnesota	2:1	2.0%	3.1%	2.4%
Florida	3:1	2.3%	4.1%	2.9%
Ohio	3:1	2.4%	3.9%	2.9%
Vermont	5:1	1.9%	5.2%	3.0%
Massachusetts	4:1	2.1%	4.5%	2.9%
New Hampshire	50:3	1.4%	2.2%	1.7%
Mean (42 states)	—	2.1%	3.4%	2.5%

Table 3: Compactness penalty (edge-cut increase) from nesting constraint. Larger H/S ratios incur higher senate penalties because larger senate districts have more interior boundaries from house-district merging. New Hampshire (50:3 ratio) has a low penalty because the 8-spine super-regions are large and the within-super-region redistricting is relatively unconstrained.

State	Ratio	Indep. D (H)	Nest D (H)	Indep. D (S)	Nest D (S)
Washington	2:1	57	57	29	29
Minnesota	2:1	76	75	39	39
Florida	3:1	54	54	18	18
Ohio	3:1	40	40	14	14
Virginia	5:2	52	52	21	21
Wisconsin	3:1	43	42	15	15

Table 4: Partisan outcomes (Democratic seat counts) for independent vs. NestSection maps. Nesting changes outcomes in at most 1 state (Wisconsin House: 43D→42D) among the 30 states with available precinct data. The expected shift is 0.3 seats averaged across chambers and states.